New Uzbekistan University

Statement of Ethics: I agree to complete this exam without unauthorized assistance.

FULL NAME: _____ SIGNATURE: _____

ID NUMBER: _____ EXAM ROOM: M A T H _____ GROUP _____

Midterm Exam

COURSE NAME: Calculus 1 COURSE NUMBER: MATH 111

EXAMINATION DURATION: 90 MINUTES EXAM VERSION A

ADDITIONAL MATERIALS ALLOWED TO USE: No additional materials or items are allowed to be used.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST! VIOLATION OF THE RULES CAN LEAD TO A LOSS OF POINTS.

- Read and strictly follow extra instructions at the beginning of each Section.
- Put your signature and write your ID on even pages.
- Follow the University Examination Rules.

FOR INSTRUCTOR USE ONLY (DO NOT WRITE ANYTHING):

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	17	7	12	7	8	12	13	17	7	100
Score:										

SIGNATURE: ID:

1 Written Response

Please provide your responses in the spaces provided. Only fully complete answers will be eligible to receive full marks.

1. (17 points) Total points

   (a) (5 points) Find, if it exists, the limit

$$\lim_{x\to 0} \frac{2x+5}{\arctan\left(\frac{\pi}{x}\right)}.$$

   (b) (5 points) Find the following limit if it exists. If it does not exist, then justify your answer.

$$\lim_{t\to 0}\left(\frac{1}{t}-\frac{1}{t^2+t}\right)$$

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(c) (7 points) Use the Squeeze Theorem to determine the value of

$$\lim_{x\to 0}\left(x^2 \arctan\left(\frac{\pi}{x}\right)\right).$$

2. (7 points) Use the precise definition of limit to prove that

$$\lim_{x\to -2}\left(3-\frac{x}{2}\right)=4.$$

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3. (12 points) Consider the function

$$g(x) = \begin{cases} ax - 3 & \text{if } x \leq 4 \\ \frac{a|2x^3 - 8x^2|}{x-4} & \text{if } x > 4 \end{cases}$$

(a) (5 points) Knowing that g is continuous everywhere, find the value of a.

(b) (7 points) For $a = 1$ find $g'(x)$ and its domain.

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4. (7 points) Prove that the equation

$$3x^5 - 4x^2 = 3$$

has a solution over the interval $[1, 2]$.

5. (8 points) Assuming that y is defined implicitly as a function of x, by the equation

$$e^{xy} = x + y$$

find an equation of the tangent line to the graph of the curve at the point $(0, 1)$

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6. (12 points) Find $\frac{dy}{dx}$ for the following functions:

(a) (3 points) $f(x) = \sqrt{3x^2 - 5x + 7}$

(b) (3 points) $f(x) = 3x^2 \operatorname{sech}(x^3)$

(c) (6 points) $f(x) = x^{\tan(x)}$

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7. (13 points) Total points

(a) (8 points) Find the absolute minimum and maximum for $f(x) = x^{2/3}(10-x)$ over the interval $[-1, 10]$.

(b) (5 points) Let $f(x) = \dfrac{2x+3}{x+2}$. Show that f satisfies the hypotheses of the Mean Value Theorem on the interval $[-1, 2]$. Find all value(s) of c that satisfy the conclusion of the theorem.

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8. (17 points) Let

$$f(x)=\frac{2x^2+3}{3x^2-12},\quad f'(x)=\frac{-22x}{3(x^2-4)^2},\quad f''(x)=\frac{22(3x^2+4)}{3(x^2-4)^3}.$$

(a) (5 points) Find the vertical and horizontal asymptotes of f.

(b) (6 points) Find the critical number(s), intervals(s) of increasing and decreasing and points of local extrema of f, if any.

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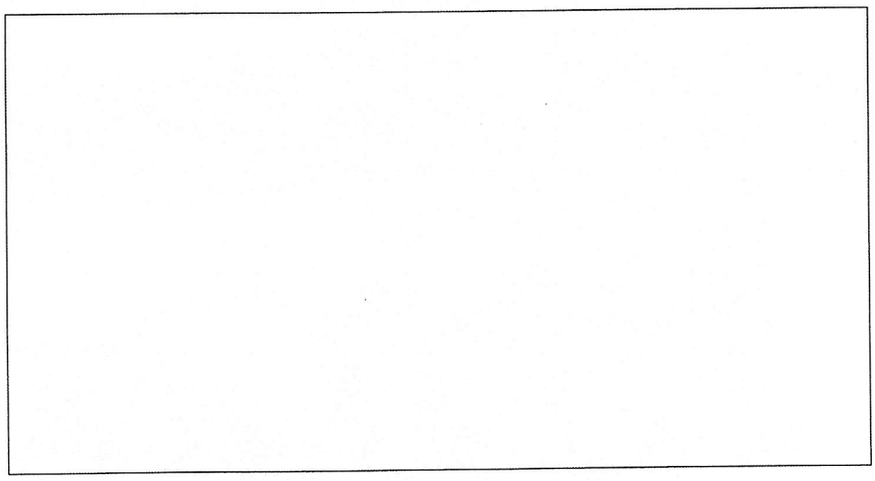
(c) (6 points) Find intervals of concavity and the point(s) of inflection of f, if any.

9. (7 points) Given that $f(x)$ is differentiable at $x = 0$, $f(0) = 1$ and satisfies the equation

$$f(x) + (f(x))^3 = 2\sin(3x)$$

Find $f'(0)$.

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